

Nitish Kumar Gupta

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PROFILE SUMMARY

AI Engineer with a strong foundation in Artificial Intelligence, Machine Learning, and Deep Learning. Experienced in developing end-to-end AI solutions using NLP and Computer Vision. Proficient in model development, optimization, and deployment using modern AI frameworks.

INTERNSHIP

Codenscious.ai

Jul 2024 – Oct 2024

Computer Vision Intern

Indore, India

- Performed data preprocessing, cleaning, annotation, and augmentation to improve model performance.
- Developed and optimized deep learning-based computer vision models, improving accuracy through hyperparameter tuning.
- Collaborated with team members to improve model efficiency and ensure smooth deployment readiness.

PROJECTS

Intelligent Question Generator (NLP)

- Developed an NLP-based question generation system using Streamlit and LLaMA to generate MCQs and other question formats efficiently.
- Designed an end-to-end pipeline integrating LLaMA and WordNet for distractor generation with adaptive difficulty levels.

Pothole Detection System

- Developed and deployed a real-time pothole detection system using deep learning, computer vision, and Raspberry Pi for efficient edge-based road monitoring.
- Integrated GPS-based location tracking with Power BI dashboards to map detected potholes and generate analytical insights for data-driven road maintenance.

Ringo Robot (Autonomous Navigation)

- Developed an autonomous robot simulation using ROS2 and Gazebo, implementing real-time navigation, path planning, and collision avoidance algorithms.
- Integrated LiDAR-based obstacle detection and ROS2 nodes to enable dynamic environment perception and efficient autonomous navigation.

SKILLS

- **Programming Languages:** Python, C, HTML, CSS
- **AI/ML:** Machine Learning, Deep Learning (CNN, RNN), NLP, Computer Vision, Generative AI
- **Frameworks & Libraries:** TensorFlow, PyTorch, Keras, OpenCV, scikit-learn
- **Data Science:** NumPy, Pandas, Matplotlib, Seaborn, SciPy
- **Tools:** Power BI, Jupyter Notebook, Git, VS Code
- **Robotics:** ROS2, Gazebo, Raspberry Pi

EDUCATION

Indore Institute of Science and Technology

Bachelor of Technology in Artificial Intelligence & Machine Learning (CGPA: 7.66)

2021 – 2025

Indore, India

CERTIFICATIONS

- Introduction to Data Science
- Python Basics for Data Science